

A Guide to Statistical Network Modeling: ERGMs, Temporal ERGMs, and SAOMs

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Abstract

Statistical network models differ from conventional regression models because relational observations are generally dependent: the probability of a tie may depend on other ties, local configurations, actor attributes, dyadic covariates, and prior network states. This guide covers three major families of models for such data: exponential-family random graph models (ERGMs) for cross-sectional networks, temporal and separable temporal ERGMs (TERGMs and STERGMs) for discrete-time network evolution, and stochastic actor-oriented models (SAOMs) for actor-driven continuous-time network dynamics observed at discrete waves. The presentation emphasizes mathematical formulation, assumptions, estimation, interpretation, diagnostics, and model choice. Curved ERGMs, geometrically weighted statistics, degeneracy, goodness of fit, formation–dissolution separability, rate and objective functions, and joint network–behavior models for distinguishing selection from influence are treated explicitly. Proofs are omitted, but definitions and inferential statements are given in sufficient detail to make clear what each model asserts probabilistically and what its parameters mean. The final synthesis stresses that these frameworks answer related but non-identical questions: ERGMs describe dependence in a network distribution, TERGMs model transitions between observed network states, STERGMs distinguish formation from persistence or dissolution, and SAOMs represent network evolution as sequences of actor-level opportunities for change.

1 Introduction

Network data consist of relational measurements among units. Let $V = \{1, \dots, n\}$ denote a fixed actor set and let Y_{ij} indicate the presence of a relation from actor i to actor j . For a binary directed network, $Y_{ij} \in \{0, 1\}$ for $i \neq j$ and usually $Y_{ii} = 0$; for an undirected network $Y_{ij} = Y_{ji}$. The random adjacency matrix is denoted by $\mathbf{Y}$, its realization by $\mathbf{y}$, and the sample space of admissible networks by $\mathcal{Y}$.

The central statistical difficulty is relational dependence. Treating dyads as independent observations is often scientifically implausible. Reciprocity makes Y_{ij} informative about Y_{ji} ; closure makes Y_{ij} related to ties through third actors; degree heterogeneity implies that many dyads incident to the same actor may share a dependence mechanism. Statistical network modeling therefore

requires probability distributions over entire graphs or stochastic processes whose states are graphs (Wasserman and Faust, 1994; Robins et al., 2007).

This guide focuses on three related traditions. Static ERGMs assign an exponential-family probability distribution to a network observed at one time. TERGMs extend this logic to transitions among networks observed at multiple times. STERGMs parameterize formation and dissolution separately (Krivitsky and Handcock, 2014). SAOMs instead posit an underlying continuous-time Markov process in which actors receive opportunities to change outgoing ties, with observed network data interpreted as snapshots of that process (Snijders, 2001; Snijders et al., 2010).

Throughout, the goal is neither to prove the underlying statistical results nor to prescribe a single model family. It is to make the stochastic object, assumptions, estimands, and interpretation of each framework explicit. A model is useful only relative to a research design: a structural statistic is not automatically a causal mechanism, and a temporal coefficient is not automatically a causal effect.

2 Static ERGMs

2.1 Foundations of Static ERGMs

2.1.1 Network questions

ERGMs are appropriate when the inferential target concerns the distribution of a network conditional on measured covariates. Typical questions ask whether ties exhibit reciprocity, closure, assortative mixing, degree concentration, or covariate-associated mixing after accounting simultaneously for other modeled features. The unit of probabilistic modeling is the graph, not an isolated dyad.

The distinction between descriptive and generative questions is essential. A descriptive network statistic such as the observed clustering coefficient summarizes one graph; an ERGM parameter instead indexes how a statistic contributes to the probability of graphs throughout $\mathcal{Y}$. Thus the inferential question is not simply whether triangles are numerous, but whether networks with additional closure are favored relative to networks otherwise comparable under the remaining model terms. The comparison is conditional and model-based. The same principle applies to homophily: an observed excess of same-category ties may reflect category sizes, degree heterogeneity, reciprocity, or closure. An ERGM can represent these mechanisms simultaneously and ask whether a mixing statistic contributes after the others have been incorporated.

It is also useful to distinguish explanation of the observed network from prediction of individual ties. ERGMs are joint probability models, and their principal inferential quantities concern dependence in the graph distribution. Dyadic predictions can be computed from conditional probabilities, but excellent classification of zeros in a sparse graph is not evidence that the dependence structure has been modeled adequately. Conversely, a scientifically useful structural model need not maximize edge-level predictive accuracy. Research questions should therefore determine which statistics are included and which out-of-sample or simulation summaries are used for assessment.

2.1.2 Network data

Let $\mathbf{Y} = (Y_{ij})$ be a random graph on V . Actor covariates may be represented by x_i , and dyadic covariates by x_{ij} . Structural zeros (inadmissible dyads), bipartite constraints, directedness, and loops specify the set $\mathcal{Y}$ of network configurations that are admissible under the model. Missing ties instead define an observation process and require an observed-data likelihood or other explicitly justified treatment; when that process is ignorable for the specified model, likelihood inference can marginalize over the unobserved dyads (Handcock and Gile, 2010).

The sample space is part of the statistical specification. For an undirected loopless graph there are $\binom{n}{2}$ potential dyads and $|\mathcal{Y}| = 2^{\binom{n}{2}}$ if no further constraints apply; for a directed loopless graph there are $n(n-1)$ potential dyads and $|\mathcal{Y}| = 2^{n(n-1)}$. Structural zeros reduce this support, while bipartite networks replace the one-mode dyad set by pairs across modes. Because the normalizing constant sums over the support, changing admissible dyads changes the probability model itself.

Network boundaries deserve the same attention as sampling units in conventional statistics. If actors outside the observed boundary can carry substantial relational traffic, degree and closure statistics computed on the induced subgraph need not correspond to those in the underlying network. Missing dyads are likewise different from observed zeros. Modern implementations can condition on observed portions or restrict the support, but the analyst must state what is regarded as random, observed, missing, or impossible. Covariates should be aligned to the same actor and dyad definitions; time-varying attributes in later sections require an additional temporal alignment.

2.1.3 Relational dependence

Let $\mathcal{D}$ denote the set of admissible ordered or unordered pairs:

$$\mathcal{D} = \begin{cases} \{(i, j) : i \neq j\}, & \text{for a directed loopless network,} \\ \{(i, j) : i < j\}, & \text{for an undirected loopless network.} \end{cases}$$

An edge-independent Bernoulli graph satisfies

$$\Pr(\mathbf{Y} = \mathbf{y}) = \prod_{(i,j) \in \mathcal{D}} p_{ij}^{y_{ij}} (1 - p_{ij})^{1-y_{ij}}.$$

Thus the individual edge variables are mutually independent. An ERGM relaxes this factorization by permitting sufficient statistics that involve several edges. Dependence is thereby represented through specified relational configurations rather than through an unspecified residual correlation.

For undirected binary networks, edge independence and dyad independence coincide because each unordered dyad contains a single binary edge variable. For directed networks they must be distinguished. Dyad independence means that the pairs (Y_{ij}, Y_{ji}) associated with distinct unordered actor pairs $\{i, j\}$ are independent, while the two directed edges within a dyad may remain dependent. A reciprocity term, for example, can therefore define a dyad-independent but not edge-independent model. Under edge independence, logistic-regression representations arise naturally for individual

binary ties; under directed dyad independence with within-dyad dependence, the corresponding factorization is instead over the four possible dyad states $(0, 0), (1, 0), (0, 1), (1, 1)$.

Dependence should not be understood as an unstructured correlation term: the sufficient statistics specify which relational configurations alter relative graph probabilities. For undirected graphs, an important special case is the Markov-graph model of Frank and Strauss (1986). Under the Frank–Strauss Markov assumption, two edge variables Y_{ij} and Y_{kl} whose actor sets are disjoint are conditionally independent given the remainder of the graph:

$$Y_{ij} \perp Y_{kl} \mid \mathbf{Y}_{-(ij),(kl)} \quad \text{whenever} \quad \{i, j\} \cap \{k, l\} = \emptyset.$$

For strictly positive distributions satisfying this local dependence structure, the associated undirected exponential-family representation involves edge, k -star, and triangle configurations. General ERGMs need not satisfy this Markov restriction and may include statistics encoding dependence over larger or nonlocal configurations. Thus the substantive content of an ERGM depends critically on the sufficient statistics chosen to represent relational organization.

2.1.4 ERGM concepts

The canonical ERGM is

$$\Pr_{\boldsymbol{\theta}}(\mathbf{Y} = \mathbf{y} \mid \mathbf{X}) = \frac{\exp\{\boldsymbol{\theta}^T \mathbf{g}(\mathbf{y}, \mathbf{X})\}}{\kappa(\boldsymbol{\theta}, \mathbf{X})}, \quad \kappa(\boldsymbol{\theta}, \mathbf{X}) = \sum_{\mathbf{y}' \in \mathcal{Y}} \exp\{\boldsymbol{\theta}^T \mathbf{g}(\mathbf{y}', \mathbf{X})\}. \quad (1)$$

Here $\mathbf{g} = (g_1, \dots, g_p)^T$ is a vector of network statistics and $\boldsymbol{\theta} \in \mathbb{R}^p$ the canonical parameter. The normalizing constant is generally intractable because $|\mathcal{Y}|$ grows exponentially with the number of possible dyads (Hunter and Handcock, 2006).

For an admissible dyad (i, j) , let $\mathbf{y}_{ij}^+$ and $\mathbf{y}_{ij}^-$ agree on every dyad except (i, j) , with $y_{ij} = 1$ and $y_{ij} = 0$, respectively. Treating the covariates $\mathbf{X}$ as fixed, define the change statistic by

$$\Delta_{ij} \mathbf{g}(\mathbf{y}, \mathbf{X}) = \mathbf{g}(\mathbf{y}_{ij}^+, \mathbf{X}) - \mathbf{g}(\mathbf{y}_{ij}^-, \mathbf{X}).$$

Because $\mathbf{y}_{ij}^+$ and $\mathbf{y}_{ij}^-$ are constructed from the same configuration of all other dyads, $\Delta_{ij} \mathbf{g}(\mathbf{y}, \mathbf{X})$ depends on $\mathbf{y}_{-ij}$ and $\mathbf{X}$, but not on the current value of y_{ij} . Then

$$\text{logit } \Pr(Y_{ij} = 1 \mid \mathbf{Y}_{-ij} = \mathbf{y}_{-ij}, \mathbf{X}) = \boldsymbol{\theta}^T \Delta_{ij} \mathbf{g}(\mathbf{y}, \mathbf{X}). \quad (2)$$

This conditional-logit identity applies when both toggle states $\mathbf{y}_{ij}^+$ and $\mathbf{y}_{ij}^-$ (i.e., the networks obtained by setting $y_{ij} = 1$ and $y_{ij} = 0$, respectively, while holding $\mathbf{y}_{-ij}$ fixed, where $\mathbf{y}_{-ij}$ denotes the states of all dyads other than (i, j)) are admissible elements of the specified network support. Under structural-zero, bipartite, fixed-degree, or other constrained supports, it applies only to dyads whose conditional support contains both 0 and 1; otherwise no binary toggle comparison is defined for that dyad.

This identity is fundamental for interpretation: a coefficient is a conditional log-odds contribution per unit change in its statistic, holding the remainder of the network fixed.

The exponential-family form immediately yields several useful identities. Treating $\mathbf{X}$ as fixed, define

$$\psi(\boldsymbol{\theta}; \mathbf{X}) = \log \kappa(\boldsymbol{\theta}, \mathbf{X}).$$

Then

$$\nabla_{\boldsymbol{\theta}} \psi(\boldsymbol{\theta}; \mathbf{X}) = \mathbb{E}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y}, \mathbf{X}) \mid \mathbf{X}\}, \quad \nabla_{\boldsymbol{\theta}}^2 \psi(\boldsymbol{\theta}; \mathbf{X}) = \text{Var}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y}, \mathbf{X}) \mid \mathbf{X}\}.$$

For a regular canonical exponential family, the second expression is also the expected Fisher information for the natural parameter $\boldsymbol{\theta}$. Hence the log-partition function is convex and the likelihood score equates observed sufficient statistics with their model expectations. These identities concern the canonical parameterization; curved ERGMs require the additional Jacobian and curvature terms discussed below. To simplify subsequent notation, fixed covariates will occasionally be suppressed when no ambiguity can result.

For two networks $\mathbf{y}_a, \mathbf{y}_b \in \mathcal{Y}$ under the same fixed covariates $\mathbf{X}$, their probability ratio is

$$\frac{\Pr_{\boldsymbol{\theta}}(\mathbf{Y} = \mathbf{y}_a \mid \mathbf{X})}{\Pr_{\boldsymbol{\theta}}(\mathbf{Y} = \mathbf{y}_b \mid \mathbf{X})} = \exp\left[\boldsymbol{\theta}^\top \{\mathbf{g}(\mathbf{y}_a, \mathbf{X}) - \mathbf{g}(\mathbf{y}_b, \mathbf{X})\}\right].$$

The intractable normalizing constant cancels. This ratio emphasizes that parameters assign relative weight to differences in sufficient statistics. The familiar conditional-logit expression is the special case in which y_a and y_b differ by one dyad. Importantly, it is a conditional statement given the rest of the graph; it is not the marginal probability of a randomly selected tie.

2.1.5 Covariates

Actor and dyadic covariates enter through network statistics. For a nodal attribute x_i , sender and receiver statistics in a directed graph may be

$$g_{\text{send}}(\mathbf{y}, x) = \sum_{i \neq j} y_{ij} x_i, \quad g_{\text{recv}}(\mathbf{y}, x) = \sum_{i \neq j} y_{ij} x_j.$$

A homophily statistic for categorical x is

$$g_{\text{match}}(\mathbf{y}, x) = \sum_{i < j} y_{ij} \mathbf{1}(x_i = x_j)$$

for an undirected network. These coefficients remain conditional on all other terms; a positive matching coefficient does not by itself identify a causal preference for similarity.

For categorical attributes, a full mixing matrix can be more informative than a single matching term. If $x_i \in \{1, \dots, K\}$, define $g_{ab}(\mathbf{y}) = \sum_{ij} y_{ij} \mathbf{1}(x_i = a, x_j = b)$. Subject to an identifiable reference coding, the corresponding parameters describe differential propensities for ties between categories. For continuous attributes, absolute difference, squared difference, product, or similarity

transformations encode distinct hypotheses and should not be treated as interchangeable notions of homophily.

Dyadic covariates w_{ij} enter naturally through $\sum_{ij} y_{ij} w_{ij}$. Geographic distance, organizational co-membership, prior interaction, and exogenous opportunity structures are common examples. Their interpretation remains conditional on endogenous network terms. This is important because omission of strong structural dependence can make a covariate term proxy for topology, while omission of relevant covariates can make a structural term proxy for unmeasured opportunity. Statistical adjustment does not by itself settle causal direction.

2.1.6 Structural terms

The edge count $g_E(\mathbf{y}) = \sum_{i < j} y_{ij}$ controls baseline density in an undirected graph. Directed reciprocity can be represented by

$$g_M(\mathbf{y}) = \sum_{i < j} y_{ij} y_{ji}.$$

A triangle statistic is

$$g_\Delta(\mathbf{y}) = \sum_{i < j < k} y_{ij} y_{ik} y_{jk}$$

for an undirected graph. Star statistics describe degree concentration. Because simple triangle and high-order star counts can create severe dependence and unstable likelihood geometry, modern ERGMs often use geometrically weighted summaries rather than raw configuration counts (Snijders et al., 2006; Hunter, 2007).

Raw counts often have highly unequal scales and induce strong interactions. In an undirected graph, toggling ij changes the triangle count by the number of common neighbors of i and j ; thus a positive triangle coefficient makes closure increasingly attractive where many shared partners already exist. This positive feedback helps explain why simple edge-triangle models can concentrate probability on nearly empty or nearly complete graphs.

Degree-related terms represent another central source of dependence. A k -star count is $\sum_i \binom{d_i}{k}$, so its change statistic grows rapidly with degree. Alternating or geometrically weighted star statistics damp this growth by combining star orders with decreasing weights. Analogously, geometrically weighted edgewise shared partners summarize closure over the distribution of shared-partner counts. These terms are not merely computational tricks: their decay parameters encode how quickly additional partners contribute diminishing increments to the statistic.

2.1.7 Estimation

Treating $\mathbf{X}$ as fixed, the log-likelihood is

$$\ell(\boldsymbol{\theta}; \mathbf{y}, \mathbf{X}) = \boldsymbol{\theta}^\top \mathbf{g}(\mathbf{y}, \mathbf{X}) - \psi(\boldsymbol{\theta}; \mathbf{X}),$$

with score

$$U(\boldsymbol{\theta}) = \mathbf{g}(\mathbf{y}, \mathbf{X}) - \mathbb{E}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y}, \mathbf{X}) \mid \mathbf{X}\}.$$

Thus an interior maximum-likelihood estimate in a regular canonical ERGM satisfies the moment equation

$$\mathbf{g}(\mathbf{y}, \mathbf{X}) = \mathbb{E}_{\hat{\boldsymbol{\theta}}}\{\mathbf{g}(\mathbf{Y}, \mathbf{X}) \mid \mathbf{X}\}.$$

The difficulty is computational: except for simple edge- or dyad-independent specifications, the normalizing constant and the corresponding expectations cannot be enumerated over the graph space.

Monte Carlo maximum likelihood replaces the intractable likelihood ratio by a Monte Carlo approximation based on networks simulated from a reference parameter value (Geyer and Thompson, 1992). Iterative MCMLE updates the reference distribution as estimation proceeds, so reliable inference requires adequate MCMC exploration as well as numerical optimization. Monte Carlo error is therefore part of the inferential calculation rather than merely an implementation detail.

Maximum pseudolikelihood multiplies the full conditional probabilities of individual edge variables. This yields informative binary comparisons only where both toggle states are feasible. Under global constraints such as a fixed edge count or fixed degree sequence, one-dyad conditionals can be deterministic, so ordinary dyadwise pseudolikelihood may provide no information about the parameters. For edge-independent binary ERGMs this factorization is the likelihood itself, so MPLE and MLE coincide. Under relational dependence, however, pseudolikelihood is only an approximation to the joint likelihood, and treating its logistic-regression representation as if the conditional observations were independent can substantially understate uncertainty (van Duijn et al., 2009). MPLE can nevertheless provide useful starting values and, in some applications, a computationally inexpensive preliminary fit.

2.1.8 Interpretation

Equation (2) gives the local interpretation. If toggling (i, j) increases statistic g_k by one while all other change statistics remain fixed, then the conditional odds are multiplied by $\exp(\theta_k)$. For statistics whose toggle change is not one, interpretation must use the actual change statistic. Marginal probabilities cannot in general be read directly from coefficients because the model is jointly dependent.

2.2 Curved ERGMs, Fit, and Degeneracy

2.2.1 Curved specifications

A curved exponential family has

$$\Pr_{\boldsymbol{\theta}}(\mathbf{Y} = \mathbf{y}) = \frac{\exp\{\boldsymbol{\eta}(\boldsymbol{\theta})^T \mathbf{g}(\mathbf{y})\}}{\kappa\{\boldsymbol{\eta}(\boldsymbol{\theta})\}},$$

where the natural parameter $\eta(\boldsymbol{\theta})$ is a nonlinear function of a lower-dimensional parameter $\boldsymbol{\theta}$. Curved specifications permit families of related configurations to be represented parsimoniously and can regularize otherwise unstable high-order dependence (Hunter, 2007).

Curvature means that the natural-parameter vector lies on a lower-dimensional manifold. Let $\eta(\boldsymbol{\theta}) \in \mathbb{R}^q$, with $q > p = \dim(\boldsymbol{\theta})$, and write

$$D\eta(\boldsymbol{\theta}) = \frac{\partial \eta(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^\top}$$

for its $q \times p$ Jacobian. Suppressing fixed covariates for clarity, the score is

$$U(\boldsymbol{\theta}) = D\eta(\boldsymbol{\theta})^\top [\mathbf{g}(\mathbf{y}) - \mathbb{E}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y})\}].$$

The expected Fisher information for $\boldsymbol{\theta}$ is

$$\mathcal{I}(\boldsymbol{\theta}) = D\eta(\boldsymbol{\theta})^\top \text{Var}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y})\} D\eta(\boldsymbol{\theta}).$$

The observed negative Hessian is

$$\mathcal{J}(\boldsymbol{\theta}) = D\eta(\boldsymbol{\theta})^\top \text{Var}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y})\} D\eta(\boldsymbol{\theta}) - \sum_{k=1}^q [g_k(\mathbf{y}) - \mathbb{E}_{\boldsymbol{\theta}}\{g_k(\mathbf{Y})\}] \nabla^2 \eta_k(\boldsymbol{\theta}).$$

At a curved-family MLE, the score condition implies only

$$D\eta(\hat{\boldsymbol{\theta}})^\top [\mathbf{g}(\mathbf{y}) - \mathbb{E}_{\hat{\boldsymbol{\theta}}}\{\mathbf{g}(\mathbf{Y})\}] = 0;$$

the discrepancy vector itself need not vanish componentwise. Consequently, the second-derivative term in the observed information need not disappear at the MLE. This distinguishes expected Fisher information from observed information in a curved parameterization.

In practice, standard-error calculations may be based on either the inverse observed information or the inverse expected Fisher information, depending on the estimation procedure and software implementation. For curved models these need not coincide even at the MLE, so reported standard errors should be interpreted with reference to the information matrix actually used by the fitting software.

Their importance in network analysis comes from representing whole sequences of local configurations with a few interpretable parameters. Instead of estimating a separate coefficient for every k -star or shared-partner count, a decay parameter controls how contributions change with k . This reduces dimension and often produces more realistic dependence than unconstrained collections of high-order counts.

Mean-value parameterization. Besides the canonical parameter, an exponential family may be described through its mean-value parameter

$$\boldsymbol{\mu}(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y})\}.$$

For a regular minimal exponential family, the mapping from the canonical parameter to the relative interior of the convex support of the sufficient statistics is one-to-one. The canonical parameter describes changes in log relative probabilities, whereas the mean-value parameter describes the corresponding expected network statistics. This distinction is useful for understanding both interpretation and likelihood geometry: movement of $\boldsymbol{\mu}(\boldsymbol{\theta})$ toward the relative boundary of the convex support can signal regions in which fitted distributions become highly concentrated. For curved ERGMs, however, the attainable mean-value vectors are restricted by the lower-dimensional parameterization, so the full-family correspondence should not be transferred mechanically to the curved family.

2.2.2 Geometrically weighted terms

Geometrically weighted degree (GWD), edgewise shared-partner (GWESP), and dyadwise shared-partner (GWDSP) statistics combine configuration counts so that the incremental contribution of additional configurations decreases geometrically. Under the convention of Hunter (2007), the geometrically weighted degree statistic is

$$\text{GWD}(\mathbf{y}; \alpha) = e^{\alpha} \sum_{k=1}^{n-1} \left\{ 1 - (1 - e^{-\alpha})^k \right\} D_k(\mathbf{y}),$$

where $D_k(\mathbf{y})$ is the number of actors of degree k . The weight attached to degree k increases with k and is bounded by e^{α} ; what decreases geometrically is the increment between successive weights.

The corresponding shared-partner statistics use the same weighting form:

$$\text{GWESP}(\mathbf{y}; \alpha) = e^{\alpha} \sum_{k=1}^{n-2} \left\{ 1 - (1 - e^{-\alpha})^k \right\} EP_k(\mathbf{y}),$$

and

$$\text{GWDSP}(\mathbf{y}; \alpha) = e^{\alpha} \sum_{k=1}^{n-2} \left\{ 1 - (1 - e^{-\alpha})^k \right\} DP_k(\mathbf{y}),$$

where $EP_k(\mathbf{y})$ is the number of edges whose endpoints have exactly k shared partners and $DP_k(\mathbf{y})$ is the number of dyads whose endpoints have exactly k shared partners. Alternative software parameterizations exist, so the decay convention should always be stated explicitly.

For the weighting formulas above, take $\alpha > 0$, with $\alpha = 0$ understood as a limiting case. Smaller decay values produce faster saturation of the weights. Fixing decay is a substantive restriction: with decay fixed, the geometrically weighted statistic is a fixed canonical ERGM term; estimating decay

produces a curved parameterization. In practice, geometrically weighted terms are often preferred to raw triangles and stars because they allow closure or degree concentration without assigning an undamped reward to every additional configuration.

2.2.3 Degree distributions

Degree heterogeneity is often a first-order feature of observed networks. Raw k -star terms can model it but may generate unstable specifications. Geometrically weighted degree terms summarize the full degree distribution with diminishing marginal contributions and can distinguish degree concentration from closure mechanisms.

Degree is simultaneously an outcome to be fitted and a mechanism that shapes opportunities for further ties. An edges-only ERGM implies a binomial degree distribution under homogeneous dyad independence, which is often far too narrow for social networks. Degree terms permit overdispersion, popularity, activity, or isolates to be represented explicitly. Directed networks require separate attention to indegree and outdegree.

A common mistake is to include many exact degree indicators without considering their dependence with edges, stars, and closure. The resulting sufficient statistics may be nearly collinear or force the model toward a narrow boundary region. Geometrically weighted degree terms provide a lower-dimensional alternative. GOF should nevertheless compare the entire simulated degree distribution with the observed one, because a statistically significant degree-related coefficient does not guarantee adequate reproduction of its shape.

2.2.4 Network closure

GWESP summarizes edgewise shared partners and is commonly used to represent transitive closure in undirected networks. Its change statistic depends on how many shared partners a toggled edge possesses and on how the toggle changes shared-partner counts of neighboring edges. A positive coefficient therefore expresses a conditional tendency toward closure under the specified decay, not simply “more triangles.”

Closure can be represented in several non-equivalent ways. Triangle counts reward every closed triple; edgewise shared-partner terms condition on an existing edge and summarize how many partners its endpoints share; dyadwise shared-partner terms include both tied and untied dyads. In directed networks, transitive and cyclic triads distinguish different orientations. The statistic should therefore correspond to the hypothesized relational process rather than to a generic desire to “control for clustering.”

Closure terms interact strongly with density and degree. A graph with more edges mechanically has more opportunities for triangles, while high-degree actors create many two-paths that can close. Conditional interpretation asks whether closure is favored after these other modeled tendencies are taken into account. Simulation-based GOF can then assess whether local closure terms reproduce global clustering and shared-partner distributions.

2.2.5 Degeneracy

Degeneracy concerns the shape of the fitted or candidate graph distribution rather than the mere failure of an estimation algorithm. In a practically degenerate region, probability may concentrate on a small collection of structurally extreme graphs—often graphs much sparser or denser than the observation—so that modest parameter changes produce very large changes in expected network statistics (Schweinberger, 2011). In mean-value terms,

$$\mu(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y})\},$$

such regions are associated with highly sensitive movement of the expected sufficient statistics toward extreme parts of the convex support.

Simple edge–triangle and related high-order specifications provide important examples: positive feedback between density and closure can produce sharp transitions between sparse and dense graph regimes and near-degenerate behavior (Handcock, 2003; Chatterjee and Diaconis, 2013). In simulation, this may appear as chains moving toward extreme modes or switching only rarely between separated regions of graph space. Such behavior can cause MCMLE failure, but the computational failure is evidence about the underlying distribution rather than the definition of degeneracy itself. Conversely, poor mixing caused by an inefficient sampler or an insufficient simulation length does not by itself establish degeneracy.

MLE existence is related to, but distinct from, degeneracy. Let

$$\mathcal{C} = \text{conv}\{\mathbf{g}(\mathbf{y}) : \mathbf{y} \in \mathcal{Y}\}$$

denote the convex support, namely the convex hull of the finite set of attainable sufficient-statistic vectors. For a minimal full-rank canonical exponential family with finite support, a finite MLE exists if and only if

$$\mathbf{g}(\mathbf{y}_{\text{obs}}) \in \text{relint}(\mathcal{C}),$$

and, when it exists, it is unique (Barndorff-Nielsen, 1978). Minimality also yields strict convexity of the log-partition function in the canonical parameter. If the observed statistic lies on the relative boundary, the ordinary finite MLE does not exist and likelihood maximization may instead approach an extended-family limit. Curved and nonminimal ERGMs require additional qualifications because the natural parameter is restricted to a lower-dimensional or redundant parameter space.

Finite-MLE nonexistence should not be conflated with degeneracy, multimodality, poor MCMC mixing, or failure of an MCMLE algorithm. These phenomena can interact, especially near the boundary of the convex support, but they are mathematically distinct. A specification may possess a finite MLE yet generate a highly concentrated or multimodal distribution, and an otherwise well-defined model may be difficult to estimate because the Monte Carlo chain explores its support poorly.

2.2.6 Convergence

MCMLE convergence requires more than an optimizer reporting termination. Monte Carlo error should be controlled, chains should mix adequately, and repeated estimation should be stable. For a canonical ERGM with a finite interior MLE, Monte Carlo estimates of the expected sufficient statistics should agree with the observed statistics within numerical precision. For a curved ERGM, the relevant condition is a near-zero projected score $D\eta(\boldsymbol{\theta})^\top[\mathbf{g}(\mathbf{y}) - \hat{\mathbb{E}}_{\boldsymbol{\theta}}\{\mathbf{g}(\mathbf{Y})\}]$, assessed relative to its Monte Carlo uncertainty; componentwise agreement of all sufficient statistics is not required. Neither condition alone establishes goodness of fit.

A practical convergence assessment should separate three questions: whether the MCMC sampler mixes for a fixed parameter, whether the likelihood algorithm has reached a stationary estimate, and whether the fitted distribution is scientifically adequate. Trace plots and effective exploration of sufficient statistics address the first; small estimating-equation discrepancies and stable repeated fits address the second; GOF addresses the third.

Monte Carlo diagnostics are particularly important near degeneracy. Poor exploration of the fitted distribution or unstable estimation of the relevant score may not be resolved simply by increasing the number of iterations. The response should depend on the diagnosis: sampler changes address computational problems, whereas an inadequate or near-degenerate specification may require different model terms. Reporting only optimizer convergence or coefficient tables conceals this distinction and is insufficient for reproducible ERGM inference.

2.2.7 Goodness of fit

Goodness of fit (GOF) assesses features not forced to equal their observed values by estimation. Draw $\mathbf{Y}^{(1)}, \dots, \mathbf{Y}^{(M)}$ from the fitted model and compare observed diagnostics $T(\mathbf{y})$ with their simulated reference distributions. Common choices include degree distributions, geodesic distances, shared-partner distributions, triad censuses, and mixing patterns; graphical envelopes or empirical quantiles can reveal systematic discrepancies in these features. The objective is not to require exact reproduction of every realized statistic, but to assess whether substantively important features of the observed graph are plausible under repeated draws from the fitted graph-valued distribution. Good fit does not prove the model true; it indicates compatibility of the selected diagnostics with the fitted distribution (Hunter et al., 2008).

GOF must also be distinguished from predictive validation. Reproducing auxiliary structural statistics in networks generated from a fitted static ERGM assesses generative adequacy under that fitted distribution; it is not the same as predicting withheld ties or future networks. In sparse networks, for example, high classification accuracy can be obtained simply by predicting absent ties and therefore provides little evidence of structural adequacy. Model assessment should be aligned with the inferential purpose: structural simulation for generative fit, held-out prediction for predictive questions, and separate causal assumptions for causal questions.

2.2.8 Model comparison

For nested ERGMs fitted by maximum likelihood, the likelihood-ratio statistic is

$$2\{\ell(\hat{\boldsymbol{\theta}}_1) - \ell(\hat{\boldsymbol{\theta}}_0)\}.$$

Under the usual regularity conditions, an interior nested comparison has an asymptotic chi-square reference distribution with degrees of freedom equal to the difference in the dimensions of the fitted parameter spaces. For curved ERGMs this means the difference in $\dim(\boldsymbol{\theta})$, not the difference in the dimensions of the natural-parameter vectors $\eta(\boldsymbol{\theta})$. Boundary hypotheses and other nonregular restrictions can invalidate the ordinary chi-square reference distribution.

AIC and BIC are also used,

$$\text{AIC} = -2\ell(\hat{\boldsymbol{\theta}}) + 2p, \quad \text{BIC} = -2\ell(\hat{\boldsymbol{\theta}}) + p \log N,$$

although the effective sample size N in dependent network data is not as unambiguous as in iid settings. Likelihood-based comparison additionally requires sufficiently accurate estimates of the log normalizing constants. In ERGMs these are generally obtained through Monte Carlo procedures such as path or bridge calculations, so Monte Carlo uncertainty in estimated log likelihoods should be small relative to the model differences being interpreted.

For nonnested models, information criteria, predictive assessment, and simulation-based GOF may all be informative, but they address different notions of model adequacy. Model comparison should therefore combine likelihood evidence with substantive interpretation, numerical stability, and simulation-based adequacy rather than rely on a single scalar criterion. A small improvement in an information criterion cannot compensate for a specification that fails to reproduce important network structure or whose inferential justification is incompatible with the available sampling design.

2.2.9 Replication and asymptotic regimes

BIC requires particular caution because its penalty depends on an asymptotic information scale. In network settings there is no universally valid choice of N . The relevant asymptotic unit may involve actors, admissible dyads under sufficiently weak dependence, independent networks, approximately independent blocks, or repeated transitions, depending on the sampling and dependence regime. Consequently, setting N mechanically equal to the number of dyads is not generally justified.

More broadly, standard errors, likelihood-ratio approximations, AIC, and BIC do not acquire their usual interpretation merely because a network contains many dyads. One observed network remains one realization from a graph-valued distribution, and general consistency or asymptotic normality results are not available for arbitrary strongly dependent ERGMs under unrestricted network growth. In particular, projectivity across network sizes is a substantive property rather than an automatic feature of exponential-family network models (Shalizi and Rinaldo, 2013; Schweinberger

et al., 2020). Inferential claims should therefore identify, at least conceptually, the replication or growth regime under which their approximation is intended.

3 Temporal ERGMs and Separable TERGMs

3.1 Temporal ERGMs for Network Change

3.1.1 Longitudinal data

Suppose networks $\mathbf{Y}^1, \dots, \mathbf{Y}^T$ are observed on a common node set at discrete times. A first-order TERGM specifies the conditional transition distribution

$$\Pr_{\theta}(\mathbf{Y}^t = \mathbf{y}^t \mid \mathbf{Y}^{t-1} = \mathbf{y}^{t-1}, \mathbf{X}^t) = \frac{\exp\{\boldsymbol{\theta}^\top \mathbf{g}(\mathbf{y}^t, \mathbf{y}^{t-1}, \mathbf{X}^t)\}}{\kappa_t(\boldsymbol{\theta}; \mathbf{y}^{t-1}, \mathbf{X}^t)}, \quad (3)$$

where

$$\kappa_t(\boldsymbol{\theta}; \mathbf{y}^{t-1}, \mathbf{X}^t) = \sum_{\mathbf{z} \in \mathcal{Y}_t} \exp\{\boldsymbol{\theta}^\top \mathbf{g}(\mathbf{z}, \mathbf{y}^{t-1}, \mathbf{X}^t)\}.$$

Here $\mathcal{Y}_t$ is the admissible network support at wave t , which may itself reflect structural zeros, actor composition, or other restrictions.

Under the first-order Markov specification, the likelihood conditional on the first observed network is

$$L(\boldsymbol{\theta} \mid \mathbf{y}^1, \dots, \mathbf{y}^T, \mathbf{X}) = \prod_{t=2}^T \Pr_{\theta}(\mathbf{Y}^t = \mathbf{y}^t \mid \mathbf{Y}^{t-1} = \mathbf{y}^{t-1}, \mathbf{X}^t).$$

This factorization is a consequence of the specified conditional transition law; it does not imply that the observed transitions constitute independent unconditional replicates. If parameters are constant across waves, the same $\boldsymbol{\theta}$ governs all modeled transitions. Period indicators, interactions, or transition-specific parameters can represent temporal heterogeneity, although richly time-varying specifications may be weakly identified when only a few transitions are observed.

The first-order Markov assumption is a modeling restriction, not a property guaranteed by the temporal data. Higher-order dependence can be represented by including $\mathbf{Y}^{t-2}, \dots, \mathbf{Y}^{t-K}$ in the transition statistics. Whether such lags are meaningful depends on the substantive time scale. If a tie can change repeatedly between observations, the observed transition summarizes the net result rather than the complete event history.

3.1.2 Research design

Temporal modeling does not repair a weak observation design. The actor set, measurement process, spacing of waves, missingness, and meaning of tie persistence must be defined. Effects are conditional on the observation interval: the same social process sampled monthly and yearly need not yield the same transition parameters.

The spacing between observation waves determines the process that can be learned. At very short intervals most dyads may be unchanged, producing overwhelming stability; at very long intervals, formation and dissolution can both occur between observations and the endpoint transition loses information about ordering. The same numerical coefficient can therefore represent different effective processes under different observation schedules.

Composition change also matters. Actor entry and exit alter the risk set and can mimic tie turnover unless represented explicitly. If covariates are measured at waves, the analyst must specify whether X^{t-1} , X^t , or an interval summary enters the transition. This temporal alignment is part of the estimand. A defensible analysis states the observation times, network boundary, missingness assumptions, risk sets, and whether the parameters are assumed homogeneous across intervals.

3.1.3 Temporal dependence

Statistics may depend jointly on current and lagged networks. For each transition $t-1 \rightarrow t$, it is useful to distinguish the four dyadic transition counts

$$N_{ab}^{(t)} = \sum_{(i,j) \in \mathcal{D}_t} \mathbf{1}(y_{ij}^{t-1} = a, y_{ij}^t = b), \quad a, b \in \{0, 1\},$$

where $\mathcal{D}_t$ is the set of admissible dyads for that transition. Thus $N_{01}^{(t)}$ counts formations, $N_{10}^{(t)}$ counts dissolutions, $N_{11}^{(t)}$ counts persistent ties, and $N_{00}^{(t)}$ counts persistent non-ties. Dyadic stability is

$$N_{00}^{(t)} + N_{11}^{(t)},$$

whereas tie persistence is specifically

$$N_{11}^{(t)}.$$

This distinction matters particularly in sparse networks, where stability can be dominated by unchanged zeros. Temporal dependence can additionally include delayed reciprocity, lagged closure, and lagged covariate effects (Hanneke et al., 2010; Leifeld et al., 2018).

A general order- K TERGM can be written

$$\Pr_{\theta}(\mathbf{Y}^t = \mathbf{y}^t \mid \mathbf{Y}^{t-1:t-K} = \mathbf{y}^{t-1:t-K}, \mathbf{X}^t) \propto \exp\left\{\boldsymbol{\theta}^\top \mathbf{g}(\mathbf{y}^t, \mathbf{y}^{t-1:t-K}, \mathbf{X}^t)\right\},$$

where

$$\mathbf{Y}^{t-1:t-K} = (\mathbf{Y}^{t-1}, \dots, \mathbf{Y}^{t-K})$$

denotes the relevant observed network history.

For an order- K specification, the conditional likelihood conditions on the first K observed networks and is

$$L_K(\boldsymbol{\theta} \mid \mathbf{y}^1, \dots, \mathbf{y}^T, \mathbf{X}) = \prod_{t=K+1}^T \Pr_{\theta}(\mathbf{Y}^t = \mathbf{y}^t \mid \mathbf{Y}^{t-1:t-K} = \mathbf{y}^{t-1:t-K}, \mathbf{X}^t).$$

Thus an order- K model contributes $T - K$ conditional transitions to the likelihood.

Statistics may depend solely on $\mathbf{y}^t$ (contemporaneous structure), solely on relations between current and lagged networks (memory), or on interactions between the two. For example, lagged reciprocity can count $y_{ij}^t y_{ji}^{t-1}$, while lagged transitivity can count current ties that close two-paths existing at the preceding wave.

These specifications clarify what “temporal dependence” means: dependence is encoded through chosen summaries of history. A first-order model can still have complex within-wave dependence because g may contain triangles or stars in $\mathbf{y}^t$. Conversely, including a lag does not automatically account for all persistence mechanisms. The analyst must distinguish contemporaneous structural dependence from dependence on previous network states.

3.1.4 Network persistence

Networks are often highly persistent. Failure to model persistence can cause structural terms to absorb inertia. Conversely, a large stability parameter may dominate the transition distribution and obscure formation mechanisms. Persistence is therefore both a substantive process and a baseline against which other temporal effects are interpreted.

Persistence also changes the meaning of contemporaneous statistics. If most existing ties survive, much of the degree and closure structure at time t is inherited from $t - 1$. A contemporaneous triangle term can then combine effects of inherited closure and newly produced closure. STERGM addresses this ambiguity by assigning separate risk sets to absent and present ties.

3.1.5 Lagged effects

A lagged statistic $g(\mathbf{y}^t, \mathbf{y}^{t-1})$ describes association between the prior state and the conditional distribution of the next state. It does not alone establish temporal causation. Lag length is determined by observation spacing, and unobserved intermediate changes remain latent.

Lagged statistics can encode delayed reciprocity, inertia, delayed closure, or covariate-by-history interactions. For example, $\sum_{ij} y_{ij}^t y_{ji}^{t-1}$ measures whether a prior incoming tie is associated with a current outgoing tie. Yet the conditional association may arise from persistent dyadic propensities rather than a behavioral response. The temporal ordering rules out some interpretations but not confounding.

Higher-order lags increase flexibility at a cost. Consecutive network snapshots are usually highly correlated, so statistics based on adjacent lags can be collinear. The limited number of conditional transitions constrains the complexity that can be supported by the data. Thus lag order should be driven by theory and observation scale rather than by unrestricted search over many memory terms.

3.1.6 Time-varying covariates

Let X^t denote covariates measured at or before transition $t - 1 \rightarrow t$. Statistics may encode sender, receiver, dyadic, or interaction effects that vary across waves. Temporal ordering must be explicit:

a covariate measured after network change cannot straightforwardly be interpreted as a predictor of that transition.

If x_i^t changes over time, a transition statistic such as $\sum_{ij} y_{ij}^t x_i^{t-1}$ uses a clearly pre-transition predictor, whereas using x_i^t conditions on a value measured at the endpoint. The latter may be legitimate for descriptive conditional modeling but is problematic if the attribute itself can respond to network change. Joint network–behavior dynamics, treated below for SAOMs, are designed for precisely this endogeneity problem.

Dyadic exposures can also vary by interval. A contact opportunity, geographic distance, or institutional co-membership may be defined for the whole interval or at one endpoint. Missing covariate histories reduce the set of transitions that can be modeled unless an explicit missing-data strategy is adopted. These choices should be reported because they define the conditioning information in the transition probability.

3.1.7 Model assumptions

Common assumptions include a fixed or appropriately handled risk set, a specified Markov order, homogeneous parameters across transitions unless modeled otherwise, and correct support constraints. The likelihood factors by the chain rule and the specified Markov restriction; this does not make the observed transitions independent replicates, because they share network states. The TERGM is a model for observed-wave transitions, not necessarily for microsteps occurring between waves.

A homogeneous TERGM assumes that the same θ governs every modeled transition. This can be relaxed with period indicators or interactions, but the resulting parameters require enough transitions and variation. Conditional on the specified history, the likelihood factorization treats transition contributions according to the assumed Markov structure; unmodeled serial dependence can invalidate uncertainty calculations even if point estimates appear stable.

The model also presumes that the statistics are evaluated on the correct support at each wave. Structural zeros, actor turnover, or changes in possible dyads alter $\mathcal{Y}_t$. Finally, the transition law is not automatically a continuous-time process. Interpreting coefficients as instantaneous formation hazards requires assumptions not supplied by a generic discrete-time TERGM.

3.1.8 Estimation

Conditional maximum likelihood for a TERGM maximizes

$$\ell(\theta) = \sum_{t=2}^T \left[\theta^\top \mathbf{g}(\mathbf{y}^t, \mathbf{y}^{t-1}, \mathbf{X}^t) - \log \kappa_t(\theta; \mathbf{y}^{t-1}, \mathbf{X}^t) \right].$$

This likelihood is conditional on the first observed network $\mathbf{y}^1$; the marginal distribution of $\mathbf{Y}^1$ is not modeled by the TERGM. As in static ERGMs, the transition-specific normalizing constants are generally intractable when the statistics induce within-network dependence. MCMC maximum likelihood therefore approximates the required expectations by simulating networks from the

conditional transition distributions; implementations of this likelihood-based approach are available through `mtergm` and `tergm`.

A computationally cheaper alternative is maximum pseudolikelihood, which replaces each transition likelihood by a product of conditional edge probabilities over admissible dyads. Bootstrap procedures can then be used to estimate uncertainty by resampling at an appropriate transition or temporal unit. In particular, the `btergm` approach combines MPLE with bootstrap inference rather than performing MCMC maximum likelihood (Leifeld et al., 2018). These procedures should not be described interchangeably: MCMLE approximates the likelihood of the specified transition model, whereas bootstrap MPLE combines a pseudolikelihood point estimator with resampling-based uncertainty.

The amount of temporal replication matters. With only a small number of observed transitions, bootstrap distributions may provide limited information about temporal heterogeneity, and treating a very large number of dyads as independent observations does not create additional independent transitions. Bayesian and other simulation-based procedures are possible alternatives, but their inferential interpretation likewise depends on the assumed transition model and available replication.

3.1.9 Interpretation

Conditional change-statistic logic remains available for an admissible dyad (i, j) when both values of Y_{ij}^t belong to its conditional support:

$$\text{logit Pr}_{\theta}(Y_{ij}^t = 1 \mid \mathbf{Y}_{-ij}^t, \mathbf{Y}^{t-1}, \mathbf{X}^t) = \boldsymbol{\theta}^\top \Delta_{ij} \mathbf{g}(\mathbf{y}^t, \mathbf{y}^{t-1}, \mathbf{X}^t).$$

As in the static ERGM, this identity does not apply to a structurally fixed dyad for which only one toggle state is admissible. Moreover, when the transition model includes persistence, the same coefficient may have different implications for ties that were absent and ties that were present at the preceding wave. This motivates an explicit separation of formation and persistence processes.

3.2 Separable TERGMs for Formation and Dissolution

3.2.1 Separability

For two successive random networks $\mathbf{Y}^{t-1}$ and $\mathbf{Y}^t$, define the formation and persistence networks

$$\mathbf{Y}^+ = \mathbf{Y}^{t-1} \cup \mathbf{Y}^t, \quad \mathbf{Y}^- = \mathbf{Y}^{t-1} \cap \mathbf{Y}^t.$$

For realizations $\mathbf{y}^{t-1}$ and $\mathbf{y}^t$, the corresponding formation and persistence networks are

$$\mathbf{y}^+ = \mathbf{y}^{t-1} \cup \mathbf{y}^t, \quad \mathbf{y}^- = \mathbf{y}^{t-1} \cap \mathbf{y}^t.$$

Thus $\mathbf{y}^+$ contains every tie present initially or at the endpoint, whereas $\mathbf{y}^-$ contains precisely those initial ties that persist to the endpoint. Given $\mathbf{y}^{t-1}$, the endpoint is reconstructed uniquely as

$$\mathbf{y}^t = \mathbf{y}^- \cup (\mathbf{y}^+ \setminus \mathbf{y}^{t-1}) = \mathbf{y}^+ \setminus (\mathbf{y}^{t-1} \setminus \mathbf{y}^-).$$

Consequently, conditional on $\mathbf{y}^{t-1}$, the mapping

$$\mathbf{y}^t \longleftrightarrow (\mathbf{y}^+, \mathbf{y}^-)$$

is a bijection between admissible endpoint networks and compatible formation–persistence pairs. The separable construction below additionally requires a Cartesian-product support for the two components, with every allowed pair reconstructing to an admissible endpoint. Global constraints that couple formation and persistence may prevent this factorization; simply imposing a joint compatibility restriction on an otherwise independent pair does not generally preserve separability.

The following conditional-independence restriction is additional to the preceding deterministic bijection and does not follow from it: a STERGM assumes that the formation and persistence components have variation-independent parameter spaces

$$(\boldsymbol{\theta}^+, \boldsymbol{\theta}^-) \in \Theta^+ \times \Theta^-,$$

and are conditionally separable given the preceding network:

$$\Pr_{\boldsymbol{\theta}^+, \boldsymbol{\theta}^-}(\mathbf{Y}^+, \mathbf{Y}^- \mid \mathbf{Y}^{t-1}) = \Pr_{\boldsymbol{\theta}^+}(\mathbf{Y}^+ \mid \mathbf{Y}^{t-1}) \Pr_{\boldsymbol{\theta}^-}(\mathbf{Y}^- \mid \mathbf{Y}^{t-1}).$$

This does not imply dyad independence within either component: each conditional distribution may itself contain endogenous ERGM dependence.

Under this construction, the induced endpoint transition distribution is a structured TERGM whose sufficient statistics decompose through conditionally separable formation and persistence components,

$$\mathbf{g} = (\mathbf{g}^+(\mathbf{y}^+, \mathbf{y}^{t-1}), \mathbf{g}^-(\mathbf{y}^-, \mathbf{y}^{t-1})).$$

The decomposition permits the mechanisms associated with creating absent ties to differ from those associated with retaining existing ties (Krivitsky and Handcock, 2014).

The gain is interpretive and statistical. Formation statistics are evaluated on dyads at risk of being created, whereas persistence statistics are evaluated on existing ties at risk of ending. A single predictor may enter both components with unrelated coefficients. Thus STERGM resembles event-history reasoning about incidence and duration while retaining network dependence within the two risk-set processes.

3.2.2 Tie formation

The formation model has support restricted to supergraphs of $\mathbf{y}^{t-1}$:

$$\Pr_{\theta^+}(\mathbf{Y}^+ = \mathbf{y}^+ \mid \mathbf{Y}^{t-1} = \mathbf{y}^{t-1}) \propto \exp\left\{(\boldsymbol{\theta}^+)^\top \mathbf{g}^+(\mathbf{y}^+, \mathbf{y}^{t-1})\right\}.$$

Only dyads absent at $t - 1$ are at risk of formation. For an admissible absent dyad (i, j) whose two formation states are both in the conditional support,

$$\log \frac{\Pr_{\theta^+}(Y_{ij}^+ = 1 \mid \mathbf{Y}_{-ij}^+, \mathbf{Y}^{t-1})}{\Pr_{\theta^+}(Y_{ij}^+ = 0 \mid \mathbf{Y}_{-ij}^+, \mathbf{Y}^{t-1})} = (\boldsymbol{\theta}^+)^\top \Delta_{ij}^+ \mathbf{g}^+.$$

The support restriction is crucial: a tie already present at $t - 1$ is not a formation opportunity.

Closure in formation can be defined using the previous network, the evolving formation network, or particular operator conventions. These alternatives correspond to different mechanisms. A formation homophily coefficient describes differential creation of same-attribute ties, not the cross-sectional prevalence of such ties, which also depends on their subsequent duration.

In `tergm`, `Form()` evaluates statistics on the formation network $\mathbf{Y}^+$, whereas `Persist()` evaluates them on the persistence network $\mathbf{Y}^-$. The `Diss()` operator negates statistics evaluated on that same persistence network; it does not generally evaluate them on the set of dissolved ties. A legacy `stergm` dissolution formula therefore maps to `Persist()`; using `Diss()` instead requires the corresponding sign convention.

3.2.3 Tie persistence and dissolution

The persistence component has support restricted to subgraphs of $\mathbf{y}^{t-1}$:

$$\Pr_{\theta^-}(\mathbf{Y}^- = \mathbf{y}^- \mid \mathbf{Y}^{t-1} = \mathbf{y}^{t-1}) \propto \exp\left\{(\boldsymbol{\theta}^-)^\top \mathbf{g}^-(\mathbf{y}^-, \mathbf{y}^{t-1})\right\}.$$

Here $\mathbf{Y}^-$ denotes the *persistence network*: a tie present in $\mathbf{Y}^-$ has survived the transition. The corresponding dissolution set is

$$\mathbf{Y}^{\text{diss}} = \mathbf{Y}^{t-1} \setminus \mathbf{Y}^-.$$

Thus a positive coefficient on a persistence statistic favors retention of the associated configuration; its verbal implication for dissolution has the opposite direction. The distinction between $\mathbf{Y}^-$ and $\mathbf{Y}^{\text{diss}}$ should be maintained explicitly rather than using “dissolution network” for both objects.

A structural persistence effect can make tie duration context-dependent. For example, a positive persistence term for embedded ties implies that ties with shared partners survive more readily, even if formation closure is weak. This distinction permits a network to display strong cross-sectional clustering because closed ties last longer rather than because they form more frequently. Such mechanisms cannot be separated from a single cross-section.

3.2.4 Process-specific effects

A nodal similarity effect may be strongly positive for formation but negligible for persistence, or vice versa. A closure effect may create new ties among actors with shared partners without necessarily prolonging existing ties. STERGM coefficients therefore correspond more directly to process-specific hypotheses than coefficients in a transition model that combines incidence and persistence.

The stationary or long-run network distribution, when it exists, reflects the balance of incidence and duration. High prevalence of a tie type can arise from high formation, long duration, or both. Consequently, cross-sectional overrepresentation does not identify which process produced it. STERGM decomposes these contributions by assigning distinct parameters to formation and persistence.

For an edges-only persistence specification, under the sparse-network and long-mean-duration conditions considered by Carnegie et al. (2015), the relationship between a cross-sectional ERGM edges coefficient and the corresponding formation coefficient can be approximated by

$$\theta_{\text{edges}}^+ \approx \theta_{\text{edges}}^{\text{cs}} - \log\{\mathbb{E}(D) - 1\}.$$

Indeed, for the edges-only persistence model, $\mathbb{E}(D) = 1 + \exp(\theta_{\text{edges}}^-)$, so that $\theta_{\text{edges}}^- = \log\{\mathbb{E}(D) - 1\}$. This special-case approximation illustrates how cross-sectional prevalence combines formation propensity with duration; it is not a general identity between cross-sectional ERGM and STERGM coefficients.

This also changes policy or intervention interpretation. A covariate associated with formation may affect opportunities for new relations, whereas a persistence effect concerns maintenance of existing relations. Even when the same statistic appears in both equations, equal numerical coefficients are not generally implied or substantively expected because the risk sets and normalizing constants differ.

3.2.5 Observation intervals

Discrete-time separability is conditional on the chosen interval. Multiple unobserved formation and dissolution events may occur between waves. If intervals vary, formation and persistence probabilities are not directly comparable without modeling duration or exposure. Continuous-time interpretations require additional assumptions.

If the underlying process is approximately continuous in time, endpoint-persistence probabilities depend on interval length. With a constant dissolution hazard μ , the probability of *uninterrupted survival* over interval Δ is $\exp(-\mu\Delta)$. This need not equal $\Pr(Y_{ij}^t = 1 \mid Y_{ij}^{t-1} = 1)$ when a tie can dissolve and reform between observations. Thus coefficients estimated from annual observations cannot generally be transported unchanged to monthly data. Formation probabilities show analogous exposure dependence.

Discrete-time STERGM can still be appropriate when the observation interval itself is the scientifically meaningful unit, but the estimand should be stated accordingly. Unequal intervals

require special care: treating them as exchangeable transitions imposes a hidden equality of per-interval probabilities despite different exposure times. Duration-dependent specifications or continuous-time alternatives may then be preferable.

3.2.6 Tie duration

Under a dyad-independent, time-homogeneous, memoryless persistence model with constant per-period dissolution probability q , define the duration D so that a tie surviving no additional observation interval has $D = 1$. Then

$$\Pr(D = d) = (1 - q)^{d-1}q, \quad d = 1, 2, \dots,$$

and

$$\mathbb{E}(D) = \frac{1}{q}.$$

Equivalently, if $p = 1 - q$ denotes the per-period persistence probability,

$$\Pr(D \geq d) = p^{d-1}.$$

For the simplest edges-only persistence model,

$$\text{logit}(p) = \theta_{\text{edges}}^-,$$

so that

$$\mathbb{E}(D) = \frac{1}{1 - p} = 1 + \exp(\theta_{\text{edges}}^-).$$

Thus the persistence edges coefficient has a direct duration interpretation only under this dyad-independent, time-homogeneous, memoryless specification.

The geometric result therefore requires constant, dyad-independent, memoryless dissolution. Endogenous persistence statistics, age dependence, time-varying covariates, or period heterogeneity generally destroy this simple geometric-duration interpretation.

Observed duration data also require attention to censoring. Ties already present at the first observed wave are left-censored with respect to their true formation time, while ties surviving through the final wave are right-censored. Empirical duration summaries and model diagnostics should not treat such observations as fully observed lifetimes.

With age dependence or network-dependent survival, durations cease to be geometric. A tie's hazard can vary as its endpoints' degrees, shared partners, or covariates change. Empirical tie-age information can therefore be a powerful diagnostic for a fitted dynamic model. Matching cross-sectional density while generating unrealistic durations is evidence that the formation-persistence balance is wrong.

3.2.7 Estimation

Conditional maximum-likelihood estimation (CMLE) exploits the STERGM formation–persistence decomposition when successive network observations are available. Given the preceding network, the formation and persistence components are fitted to the observed transition on their respective constrained supports. Each component may retain the computational difficulties of an ERGM because its normalizing constant can be intractable. CMLE can therefore require MCMC approximation.

CMLE should be distinguished from equilibrium generalized method-of-moments estimation (EGMME). EGMME can be used when the available information consists of a cross-sectional network together with temporal information such as tie-duration data, but it assumes that the observed network is a realization of a TERGM process at stationarity (equilibrium). It estimates parameters by matching equilibrium network and duration features rather than by maximizing an observed-transition likelihood. The choice therefore depends on the observation design and estimand. In either case, formation and persistence parameters refer to different risk sets and should not be interpreted as two coefficients from a single ordinary logistic regression.

3.2.8 Goodness of fit

Dynamic GOF should assess both network states and transitions. Useful transition summaries include the formation and dissolution counts

$$N_{01}^{(t)} \quad \text{and} \quad N_{10}^{(t)},$$

persistence $N_{11}^{(t)}$, overall stability $N_{00}^{(t)} + N_{11}^{(t)}$, transition-specific mixing, and, when observed appropriately, tie-age or duration distributions. State-based summaries such as degree, shared-partner, or triad distributions remain useful but cannot by themselves establish that turnover is reproduced correctly.

One-step simulation asks whether the fitted model reproduces the next observed network when conditioned on the preceding observation. Multi-step simulation instead propagates the fitted process through several transitions and can reveal accumulated discrepancies in density, turnover, mixing, or structural dependence that are difficult to see in isolated one-step checks. A model may therefore fit individual observed waves reasonably well while generating implausible temporal trajectories.

4 SAOMs

4.1 SAOMs for Actor-Driven Network Dynamics

4.1.1 Actor-oriented models

A SAOM treats network evolution between observed waves as a continuous-time Markov chain $\{\mathbf{Y}(t) : t \geq 0\}$ on the graph state space. At random opportunities, an actor may change one outgoing

tie (or make no change). Networks observed at times $t_1, \dots, t_T$ are realizations $\mathbf{Y}(t_1), \dots, \mathbf{Y}(t_T)$, while intermediate microsteps are unobserved (Snijders, 2001; Snijders et al., 2010). Throughout the SAOM section, $x \in \mathcal{Y}$ is also used as a generic realized network state when following conventional actor-oriented notation; $\mathbf{Y}(t)$ denotes the random network process.

RSiena is the principal R implementation of the SAOM framework described in this section. It provides simulation-based estimation for longitudinal network models and network–behavior coevolution models, together with tools for specification, convergence assessment, simulation, and goodness-of-fit analysis (Ripley et al., 2026). The statistical framework and its software implementation should be distinguished: SAOM refers to the model class, whereas RSiena implements that class computationally.

The continuous-time chain is characterized by transition intensities between networks differing by one admissible tie toggle. If $\mathbf{y}^{(ij)}$ is obtained from $\mathbf{y}$ by toggling y_{ij} , the off-diagonal intensity is

$$q(\mathbf{y}, \mathbf{y}^{(ij)}) = \lambda_i(\mathbf{y}; \rho) p_i(\mathbf{y}, \mathbf{y}^{(ij)}; \boldsymbol{\beta}),$$

where $\lambda_i(\mathbf{y}; \rho)$ is actor i ’s opportunity rate and $p_i(\mathbf{y}, \mathbf{y}^{(ij)}; \boldsymbol{\beta})$ is the conditional probability of choosing that tie change given an opportunity. The diagonal generator entry is

$$q(\mathbf{y}, \mathbf{y}) = - \sum_{\mathbf{y}' \neq \mathbf{y}} q(\mathbf{y}, \mathbf{y}').$$

Thus the generator records actual state changes. A no-change choice may occur at an actor opportunity, but it is a null opportunity rather than an off-diagonal jump of the continuous-time Markov chain.

The likelihood of the observed network sequence integrates over all unobserved microstep sequences connecting consecutive waves. Direct evaluation is therefore difficult. Simulation is central both to estimation and to interpretation. The microstep assumption is substantive: simultaneous coordinated changes are represented as rapid sequences of individual moves unless the model is extended. For relations whose changes are genuinely event-like rather than enduring states, this latent-process interpretation may be inappropriate.

The basic SAOM formulation therefore rests on several substantive assumptions: the latent network process is Markovian; elementary network changes occur one at a time; actors’ opportunities and choices are conditionally specified from the current state and covariates, without additional dependence on the unobserved past beyond that state; the observed network at the beginning of each observation period is conditioned upon; and parameters are constant within a period unless explicitly modeled otherwise. These assumptions concern the stochastic process rather than merely the estimation algorithm.

For undirected networks, an additional protocol is required because a tie change concerns two actors rather than a sender’s unilateral outgoing-tie decision. Actor-oriented specifications may therefore use protocols such as unilateral initiative or mutual agreement. The chosen protocol is

part of the model and changes the interpretation of the microstep process (Snijders et al., 2010).

4.1.2 Actor choices

For a directed network, let $\mathbf{y}^{(ij)}$ denote the network obtained from $\mathbf{y}$ by toggling y_{ij} , provided that this toggle is admissible, and let $\mathbf{y}^{(i0)} = \mathbf{y}$ denote no change. Conditional on actor i receiving an opportunity, define the choice set

$$\mathcal{A}_i(\mathbf{y}) = \{\mathbf{y}\} \cup \left\{ \mathbf{y}^{(ij)} : j \neq i, \mathbf{y}^{(ij)} \in \mathcal{Y} \right\}.$$

Under the usual random-utility representation, actor i associates each candidate state $\mathbf{y}' \in \mathcal{A}_i(\mathbf{y})$ with

$$U_i(\mathbf{y}, \mathbf{y}') = f_i(\mathbf{y}, \mathbf{y}'; \boldsymbol{\beta}) + \varepsilon_i(\mathbf{y}'),$$

where the $\varepsilon_i(\mathbf{y}')$ are independent standard Gumbel disturbances. The resulting conditional choice probabilities are exactly multinomial logit:

$$p_i(\mathbf{y}, \mathbf{y}'; \boldsymbol{\beta}) = \frac{\exp\{f_i(\mathbf{y}, \mathbf{y}'; \boldsymbol{\beta})\}}{\sum_{\mathbf{z} \in \mathcal{A}_i(\mathbf{y})} \exp\{f_i(\mathbf{y}, \mathbf{z}; \boldsymbol{\beta})\}}, \quad \mathbf{y}' \in \mathcal{A}_i(\mathbf{y}).$$

Consequently,

$$\frac{p_i(\mathbf{y}, \mathbf{y}'; \boldsymbol{\beta})}{p_i(\mathbf{y}, \mathbf{y}''; \boldsymbol{\beta})} = \exp\{f_i(\mathbf{y}, \mathbf{y}'; \boldsymbol{\beta}) - f_i(\mathbf{y}, \mathbf{y}''; \boldsymbol{\beta})\}.$$

The evaluation-only specification is the special case in which the objective depends on the candidate state alone, so that $f_i(\mathbf{y}, \mathbf{y}'; \boldsymbol{\beta}) = f_i(\mathbf{y}'; \boldsymbol{\beta})$.

The actor-oriented formulation is a myopic stochastic-optimization model, not a claim that empirical actors consciously calculate or maximize the specified objective function. Constraints on permissible ties modify $\mathcal{A}_i(\mathbf{y})$ and hence the choice probabilities themselves.

4.1.3 Rate functions

The rate function $\lambda_i(x; \rho)$ determines how frequently actor i receives opportunities for change. If the total rate is $\lambda_+(x) = \sum_i \lambda_i(x)$, the waiting time to the next opportunity is exponential with rate $\lambda_+(x)$, and actor i is selected with probability $\lambda_i(x)/\lambda_+(x)$. This is the waiting time between actor opportunities, not necessarily the holding time of the network-state process. Because an opportunity may result in no change, the total intensity of actual network-state departures can be smaller than $\lambda_+(x)$. Rates may vary by observation period and actor covariates. In longitudinal data it is common to allow a distinct baseline rate parameter ρ_m for the interval $(t_m, t_{m+1}]$, because different intervals may require different amounts of latent microstep activity to connect their observed endpoints.

With constant actor rates within a period, the expected number of opportunities is proportional to elapsed continuous time. Period-specific baseline rates absorb how much unobserved change is required to connect successive observations. Covariate-dependent rates can represent actors who

receive more opportunities, but rate effects should be distinguished from objective effects: the former change frequency of opportunities, the latter alter choices conditional on an opportunity.

This separation is useful because an actor can exhibit many observed changes either by acting frequently or by having objective-function incentives that often favor toggles over no change. Identification of elaborate rate heterogeneity may be weak with few waves, so standard applications often use parsimonious period rates unless theory strongly motivates actor-specific opportunity processes.

4.1.4 Objective functions

In the basic evaluation specification, the network evaluation function is

$$f_i(\mathbf{y}'; \boldsymbol{\beta}) = \sum_{k=1}^p \beta_k s_{ik}(\mathbf{y}'), \quad (4)$$

where $s_{ik}(\mathbf{y}')$ are actor-specific effects evaluated at the candidate state. The coefficient β_k changes the relative attractiveness of possible next states according to differences in s_{ik} .

More general SAOM specifications distinguish evaluation, creation, and endowment components. If actor i moves from realized network state $\mathbf{y}$ to candidate state $\mathbf{y}'$, the network objective function may be written

$$f_i(\mathbf{y}, \mathbf{y}') = \sum_k \beta_k^{\text{ev}} s_{ik}(\mathbf{y}') + \sum_k \beta_k^{\text{cr}} s_{ik}^{\text{cr}}(\mathbf{y}, \mathbf{y}') + \sum_k \beta_k^{\text{en}} s_{ik}^{\text{en}}(\mathbf{y}, \mathbf{y}').$$

Creation-function changes contribute to tie-creation moves, whereas endowment-function changes contribute to tie-dissolution moves and represent the gain or loss associated with giving up an existing tie. For a given underlying effect, evaluation, creation, and endowment are not three independently estimable roles: including all three produces perfect collinearity. One or two roles may be used when substantively justified and supported by sufficient observed change (Ripley et al., 2026).

If two available candidate states differ by Δs_{ik} on statistic s_{ik} , while their other objective-function statistics are equal, their conditional choice-odds ratio is multiplied by $\exp(\beta_k \Delta s_{ik})$. This resembles ERGM change-statistic interpretation, but the conditioning set differs fundamentally: SAOM probabilities compare actor-specific microsteps, whereas ERGM conditionals compare two network states differing in one dyad given the remainder of the graph.

4.1.5 Structural effects

Typical effects include outdegree, reciprocity, transitive triplets, three-cycles, indegree popularity, and outdegree activity. For example,

$$s_{i,\text{out}}(x) = \sum_j x_{ij}, \quad s_{i,\text{recip}}(x) = \sum_j x_{ij} x_{ji}.$$

A transitive-triplets effect can be represented as

$$s_{i,\text{trans}}(x) = \sum_{j,h} x_{ij}x_{ih}x_{hj}.$$

Because effects enter an actor’s choice function jointly, their coefficients are conditional on the other included effects.

The outdegree effect usually plays a role analogous to a density tendency: negative values discourage additional outgoing ties. Reciprocity rewards choices creating or maintaining reciprocated dyads. Transitive-triplet effects reward moves that create transitive closure, whereas three-cycle effects distinguish cyclic organization. Degree-related popularity and activity effects represent preferential attachment-like tendencies at the microstep level.

Effects can be strongly correlated. For example, reciprocity and transitive closure may both favor a candidate tie, while outdegree penalizes it. The fitted coefficient for each effect is conditional on the others. Model building should therefore be theory-led and accompanied by convergence checks; adding every available structural effect can create weak identification or unstable simulation without improving the representation of the process.

4.1.6 Actor covariates

Covariates can affect ego activity, alter attractiveness, and similarity. For scalar v_i , an ego effect may use $v_i \sum_j x_{ij}$, an alter effect $\sum_j x_{ij}v_j$, and a similarity effect $\sum_j x_{ij} \text{sim}(v_i, v_j)$. These represent different hypotheses and should not be collapsed under the generic label “covariate effect.”

For a directed relation, ego, alter, and similarity effects answer different questions. An ego coefficient asks whether actors with larger v_i tend to send more ties; an alter coefficient asks whether actors with larger v_j attract ties; similarity asks whether ties are favored when endpoints resemble one another. A single observed correlation between degree and an attribute cannot distinguish these mechanisms.

Interactions with structural effects can represent heterogeneous network mechanisms, such as stronger reciprocity for one category or stronger closure among actors with particular attributes. Such extensions increase interpretive complexity and should be justified by the research question. Time-varying covariates require values defined over the interval or at observation waves; if the attribute is itself endogenous, a coevolution model is more coherent than treating it as fixed.

4.1.7 Estimation

SAOMs are commonly estimated by simulation-based method of moments. Suppose networks are observed at times $t_1 < \dots < t_T$. For each observation interval $m = 1, \dots, T - 1$, let $\mathbf{S}_m$ denote a vector of statistics associated with the transition from t_m to t_{m+1} , and let $\mathbf{s}_m^{\text{obs}}$ denote its observed

value. The estimating equations have the period-specific form

$$\sum_{m=1}^{T-1} \mathbb{E}_{\rho, \beta} [\mathbf{S}_m \mid \mathbf{Y}(t_m) = \mathbf{y}(t_m)] = \sum_{m=1}^{T-1} \mathbf{s}_m^{\text{obs}}.$$

Thus each simulated transition is conditioned on its own observed period start; the procedure does not treat all periods as realizations from one unconditional stationary distribution.

Because the intermediate microstep paths are latent, the expectations are approximated by simulation. Robbins–Monro stochastic approximation updates the parameter vector using discrepancies between simulated and observed statistics until the estimating equations are approximately satisfied (Snijders, 2001; Ripley et al., 2026).

Let

$$D = \left. \frac{\partial \mathbb{E}_{\boldsymbol{\theta}}(\mathbf{S})}{\partial \boldsymbol{\theta}^{\top}} \right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}$$

denote the sensitivity matrix of the expected estimating statistics, evaluated at $\hat{\boldsymbol{\theta}}$, and let

$$\Sigma = \text{Var}_{\hat{\boldsymbol{\theta}}}(\mathbf{S})$$

denote the covariance matrix of those statistics under the fitted model. With this orientation of D , the usual sandwich approximation has the form

$$\widehat{\text{Cov}}(\hat{\boldsymbol{\theta}}) \approx D^{-1} \Sigma D^{-\top}.$$

Weak sensitivity, strong dependence among statistics, or insufficient network change can therefore produce large standard errors and unstable estimation even when the stochastic-approximation algorithm itself appears to converge.

4.1.8 Diagnostics

Convergence diagnostics assess whether the simulation-based estimating equations have been solved with adequate numerical accuracy. Effect-specific convergence t -ratios compare residual discrepancies between simulated and target statistics with their Monte Carlo variation. In standard RSiena practice, absolute convergence t -ratios below approximately 0.10 for individual effects and an overall maximum convergence ratio below approximately 0.25 are commonly used as practical convergence criteria (Ripley et al., 2026). These are numerical diagnostics, not tests of substantive model adequacy.

Post-estimation goodness of fit should therefore be assessed separately. Simulation-based GOF compares observed auxiliary statistics not directly targeted by estimation with their distributions under the fitted model. Degree distributions, triad census statistics, geodesic distances, and behavior distributions in coevolution models are common choices. In RSiena, such assessments are implemented through goodness-of-fit procedures that compare observed auxiliary statistics with

their simulated distributions, including multivariate assessment based on Mahalanobis distance (Lospinoso and Snijders, 2019; Ripley et al., 2026). Repeated estimation from different starting values can also reveal Monte Carlo instability.

For longitudinal network data, the amount of observed change itself should be examined before fitting a SAOM. Jaccard similarity between consecutive waves is commonly used as a descriptive measure of network stability. As a practical guideline, values around 0.3 or higher are often regarded as providing sufficient continuity between successive observations for SAOM analysis (Ripley et al., 2026). This is not a formal acceptance threshold: substantially lower similarity may make the assumption of gradual latent microsteps between waves implausible, whereas extremely little turnover provides weak information about change mechanisms. Jaccard similarity should therefore be treated as a design diagnostic rather than a model-validity criterion.

4.2 SAOMs for Selection and Influence

4.2.1 Network–behavior coevolution

Let $\mathbf{Y}(t)$ be the network and $Z(t) = (Z_1(t), \dots, Z_n(t))$ an actor behavior variable. A coevolution SAOM models $(\mathbf{Y}(t), Z(t))$ jointly, with separate opportunities and objective functions for network and behavior changes. This permits network selection and behavioral influence to be represented within one stochastic system (Steglich et al., 2010).

Let the joint state be $W(t) = (\mathbf{Y}(t), Z(t))$. Separate rate functions determine whether the next opportunity concerns a network move or a behavior move, and separate objective functions evaluate the corresponding local alternatives. Conditional on the current joint state, the process is Markov; dependence between network and behavior enters because each objective function may contain statistics of the other component.

This construction avoids treating the contemporaneous network as an exogenous predictor of behavior or vice versa. Nevertheless, it identifies only the mechanisms represented in the model. Unmeasured common causes, measurement error, or misspecified timing can still generate apparent selection or influence. The framework supports a more coherent decomposition than separate regressions, but it does not transform observational longitudinal data into an experiment.

4.2.2 Selection

Selection concerns whether behavior affects network choices. A similarity statistic in the network objective function can take the form

$$s_{i,\text{sim}}^{\text{net}}(x, z) = \sum_j x_{ij} \text{sim}(z_i, z_j).$$

For a quantitative behavior, a commonly used similarity transformation is

$$\text{sim}(z_i, z_j) = 1 - \frac{|z_i - z_j|}{\Delta_Z},$$

where Δ_Z is the observed range of the behavior. This transformation requires $\Delta_Z > 0$; when the behavior has no observed variation, the normalized similarity statistic is undefined and similarity-based selection is not empirically estimable. Note that in RSiena implementations the similarity statistic is typically centered by its observed mean. Average-similarity, total-similarity, and alter-based effects are distinct specifications and should not be interpreted interchangeably.

A positive coefficient means that, conditional on the modeled network process, potential tie changes leading to greater behavioral similarity receive greater relative choice probability.

Selection can involve similarity, ego behavior, alter behavior, or interactions. Similarity effects ask whether actors preferentially create or maintain ties to behaviorally similar alters after controlling for structural network tendencies. If behavior is ordinal or continuous, the similarity transformation should be defined explicitly because scaling changes the magnitude of its change statistics.

Selection is estimated from occasions on which network composition changes relative to the behavior distribution. If there is almost no network turnover, the data provide little information about which potential alters would have been chosen. This design requirement is separate from sample size in actors: a large but nearly frozen network can be less informative about selection than a smaller network with meaningful turnover.

4.2.3 Homophily

Observed similarity among connected actors is not itself evidence that selection generated the similarity. Homophily may arise because actors select similar alters, because connected actors influence one another, because both respond to shared environments, or because of combinations of these processes. Longitudinal coevolution models are designed to separate modeled selection and influence tendencies, conditional on their assumptions and included effects (Steglich et al., 2010).

Homophily is an observed pattern; selection is one process capable of generating it. Influence can increase similarity after ties form, while contextual exposure can make both tie choice and behavior similar. A coevolution model attempts to allocate longitudinal changes among specified mechanisms by using their different temporal signatures.

Accordingly, reporting a positive selection-similarity parameter as “homophily causes ties” overstates the model. The defensible statement is conditional: among modeled microsteps, choices leading to greater similarity are favored, controlling for included network and actor effects. Robust interpretation examines whether this tendency persists under alternative structural specifications and whether simulations reproduce observed mixing.

4.2.4 Social influence

Influence enters the behavior objective function. A positive average-similarity effect favors moves that increase similarity to alters. An average-alter effect instead uses a product of ego behavior and mean alter behavior, with the applicable centering conventions, and must be interpreted jointly with behavioral shape effects. If $\bar{z}_i^N(x, z)$ denotes mean neighbor behavior, a distinct, illustrative

statistic that directly rewards proximity to that mean is

$$s_{i,\text{infl}}^{\text{beh}}(x, z) = -\{z_i - \bar{z}_i^N(x, z)\}^2.$$

A positive coefficient favors microsteps reducing distance from neighbors, conditional on other behavioral effects.

This stylized expression requires a convention when actor i has no relevant neighbors, because $\bar{z}_i^N(x, z)$ is then undefined. An implementation must specify how isolates are handled. Moreover, this quadratic distance-to-neighbor-mean statistic is illustrative rather than identical to every RSiena influence effect: average similarity, average alter, and total alter statistics encode different behavioral mechanisms.

Influence statistics differ according to the hypothesized reference group. Average alter behavior, total alter behavior, average similarity, and incoming versus outgoing neighborhoods define different exposure functions. In directed networks, who is considered influential must be specified through tie orientation. Centering and scaling of behavior also affect objective-function statistics.

The meaning of a positive influence parameter depends on the chosen statistic. For the illustrative distance-based effect above, it favors moves toward the neighbor mean; for an average-alter effect, it changes the behavioral gradient in interaction with the alter mean and must be interpreted jointly with shape effects. It does not mean that every actor moves toward every neighbor, nor does it directly equal a regression slope. The realized amount of convergence depends on opportunity rates, competing shape effects, network evolution, and the distribution of available behavior states.

4.2.5 Behavioral change

For ordinal behavior, permitted microsteps often change z_i by one category upward or downward, plus no change. Behavior rate functions determine opportunity frequency, while objective functions can include linear and quadratic shape effects to represent baseline distributional tendencies alongside peer influence and covariates.

For an ordinal behavior with states $1, \dots, K$, interior actors typically have candidate moves $z_i - 1$, z_i , and $z_i + 1$, while boundary states have fewer alternatives. Linear and quadratic shape effects regulate baseline tendencies toward high, low, central, or extreme values. Covariates and influence terms then alter these local choice probabilities.

Because changes occur one unit at a time in the latent process, large observed wave-to-wave differences are represented by multiple unobserved microsteps. This is reasonable only when the measured behavior can plausibly evolve in that manner. Highly episodic or irreversible outcomes may require a different event-history or state-transition model.

4.2.6 Joint modeling

The joint process alternates stochastic network and behavior opportunities. Network changes depend on the current behavior state, and behavior changes depend on the current network state. This

mutual conditioning is what permits selection and influence to be represented simultaneously rather than estimated in separate regressions that condition on endogenous outcomes.

The joint model distinguishes modeled selection and influence through their different implications for the joint distribution of observed network and behavior changes. Intermediate microstep paths and their ordering are unobserved and are integrated over by simulation during estimation. A network move can alter subsequent behavioral exposure, while a behavior move can alter subsequent network-choice incentives, but the particular latent ordering generated in any simulation is not itself observed identifying information. Separate models that freeze one process while estimating the other generally therefore represent a different stochastic system.

Model specification should include baseline shape and structural effects sufficient to prevent selection and influence terms from absorbing unrelated tendencies. For example, influence can be confounded with regression toward the behavioral mean if shape effects are omitted, while selection similarity can proxy for network closure if structurally similar actors also have similar behavior.

4.2.7 Identifiability

Selection and influence can be difficult to distinguish when there are few waves, little behavioral change, little network turnover, strong collinearity among effects, or insufficient variation in exposure. Statistical identification is model- and design-dependent. Including both mechanisms does not guarantee that data contain enough information to estimate them precisely.

Identification requires variation in both processes and sufficient temporal resolution to distinguish their signatures. If all actors retain the same behavior, influence is not estimable; if ties never change, selection is not estimable. If similarity is almost perfectly aligned with another covariate or structural opportunity, parameters may be statistically separable only with very large uncertainty.

Practical diagnostics include large standard errors, high parameter correlations, unstable estimates across runs, and failure of moment convergence. Design-based remedies are more powerful than algorithmic ones: additional waves, shorter intervals, richer behavioral variation, and measurement of common causes can provide information that no optimizer can manufacture. Claims separating selection from influence should therefore report both model assumptions and the amount of observed change supporting each mechanism.

Three claims should therefore be kept distinct. Model identification concerns whether different parameter values imply distinguishable distributions under the specified SAOM. Statistical discrimination concerns whether the observed longitudinal data contain enough information to estimate selection and influence separately with useful precision. Causal identification of peer influence is stronger still: neither model identification nor statistical discrimination by itself rules out unmeasured common causes, contextual confounding, measurement error, or misspecified timing. A fitted influence parameter should therefore not automatically be interpreted as a causal peer effect.

4.2.8 Interpretation

SAOM coefficients describe relative probabilities of latent microsteps, not ordinary regression effects on observed wave-to-wave differences. A positive selection-similarity coefficient and a positive influence coefficient describe two distinct components of the fitted coevolution process. Their substantive implications are best assessed using predicted or simulated trajectories in addition to coefficient signs.

A useful interpretation strategy is simulation-based contrast. For example, one may simulate the fitted coevolution model and compare its implied trajectories with simulations in which a particular selection or influence parameter is set to zero while the remaining specification is held fixed. Such comparisons describe consequences within the fitted stochastic model; they are not causal counterfactual effects unless the additional assumptions required for causal interpretation are defensible.

Rate and objective parameters must also be interpreted separately. Rate parameters govern how frequently actors receive opportunities for change, whereas objective-function parameters govern relative choices conditional on an opportunity. A larger rate parameter therefore does not imply a stronger preference for any particular network configuration.

5 Choosing Among ERGMs, TERGMs, STERGMs, and SAOMs

The four labels refer to different stochastic constructions rather than interchangeable estimators. Table 1 summarizes the central distinctions.

Table 1: Conceptual comparison of major statistical network models.

Framework	Primary stochastic object	Temporal conception	Natural interpretation
ERGM	Distribution over graphs	None required	Conditional network configurations
TERGM	Distribution of $\mathbf{Y}^t$ conditional on network history	Discrete observed-wave transitions	Effects on conditional next-network distribution
STERGM	Formation and persistence/dissolution distributions	Discrete transitions with separable processes	Process-specific formation and survival tendencies
SAOM	Continuous-time Markov chain on network states	Latent actor-level microsteps between waves	Relative probabilities of actor choices and opportunity rates

For a single cross-section, an ERGM is appropriate when endogenous dependence is substantively relevant. For longitudinal network data, the choice among TERGM, STERGM, and SAOM should follow the research question and assumed data-generating process rather than a claim that one

framework is universally superior. TERGMs directly model observed-wave transition distributions; STERGMs distinguish formation from persistence or dissolution; SAOMs are natural when theory is expressed in actor-level opportunities or when network–behavior coevolution is central (Snijders et al., 2010; Leifeld et al., 2018).

Model comparison across these families cannot generally be reduced to comparing coefficients. Parameters condition on different choice sets and probability spaces. Even similarly named terms such as reciprocity or transitivity can represent different quantities. Comparative interpretation should therefore proceed through explicitly defined statistics, simulated implications, and research-design assumptions.

The frameworks are nevertheless related in mathematically informative special cases. Under the separable construction, the induced STERGM endpoint transition distribution is a structured temporal exponential-family model in which formation and persistence are represented by conditionally separable components. Under time-homogeneous specifications, such transition models may also possess stationary distributions, although the stationary distribution need not have the same ERGM form as either transition component. Likewise, particular SAOM specifications can have ERGM stationary distributions under restrictive conditions, including appropriate potential-function representations and rate structures (Snijders, 2001). These relationships are useful theoretically but do not make the frameworks interchangeable for longitudinal inference.

The distinction is especially important for TERGMs and SAOMs. A TERGM specifies probabilities for observed-wave transitions conditional on network history, whereas a SAOM specifies a continuous-time process of latent actor-oriented microsteps between observations. Consequently, apparently similar reciprocity, transitivity, or covariate terms need not induce the same network distribution or dependence structure. Comparative studies therefore recommend choosing between the frameworks from their stochastic assumptions and substantive theory rather than treating them as alternative computational estimators of one common model (Block et al., 2019; Leifeld and Cranmer, 2019).

The scope of this guide is deliberately narrower than statistical network modeling as a whole. The discussion concentrates on binary relational data and on ERGM, TERGM, STERGM, and SAOM formulations because these frameworks make relational dependence and longitudinal change particularly explicit. Extensions exist for valued and count-valued ties, multilayer and multiplex networks, bipartite networks, changing actor composition, multilevel network structures, and other forms of relational data. Other major modeling traditions—including latent-space models, stochastic block models, relational event models, and network autoregressive approaches—answer different questions and rely on different stochastic constructions. Their omission here defines the scope of the guide rather than implying that they are special cases of the four frameworks considered above.

6 Conclusion

Statistical network modeling begins from a simple but consequential premise: relational observations cannot generally be treated as independent. ERGMs address this by specifying exponential-family distributions over complete network configurations. Their change-statistic representation provides a rigorous conditional interpretation of structural and covariate effects, while curved terms and geometrically weighted statistics provide flexible ways to represent higher-order dependence. Estimation and interpretation must nevertheless be accompanied by attention to degeneracy, MCMC behavior, and simulation-based goodness of fit.

Longitudinal models add a second dimension of dependence. TERGMs model the conditional distribution of a network given its history, making persistence and lagged structure explicit. STERGMs sharpen this representation by separating the creation of ties from their persistence or dissolution. This distinction is important whenever the mechanisms producing relationships differ from those sustaining them.

SAOMs adopt a different temporal ontology. They represent change as a continuous-time sequence of stochastic actor opportunities between successive network observations. Rate functions govern opportunities; objective functions govern relative probabilities of possible microsteps. Joint network-behavior SAOMs then provide a principled framework for representing selection and social influence simultaneously, while making clear that their empirical separation depends on longitudinal variation and modeling assumptions.

No framework eliminates the need for research design or substantive theory. Network statistics acquire meaning from the probability model in which they are embedded, and temporal coefficients acquire meaning from assumptions about observation, risk sets, and latent change. The central practical principle is therefore to align the stochastic model with the scientific question: distributions of network structure, discrete-time transitions, formation and dissolution, or actor-driven coevolution are related targets, but they are not the same target.

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